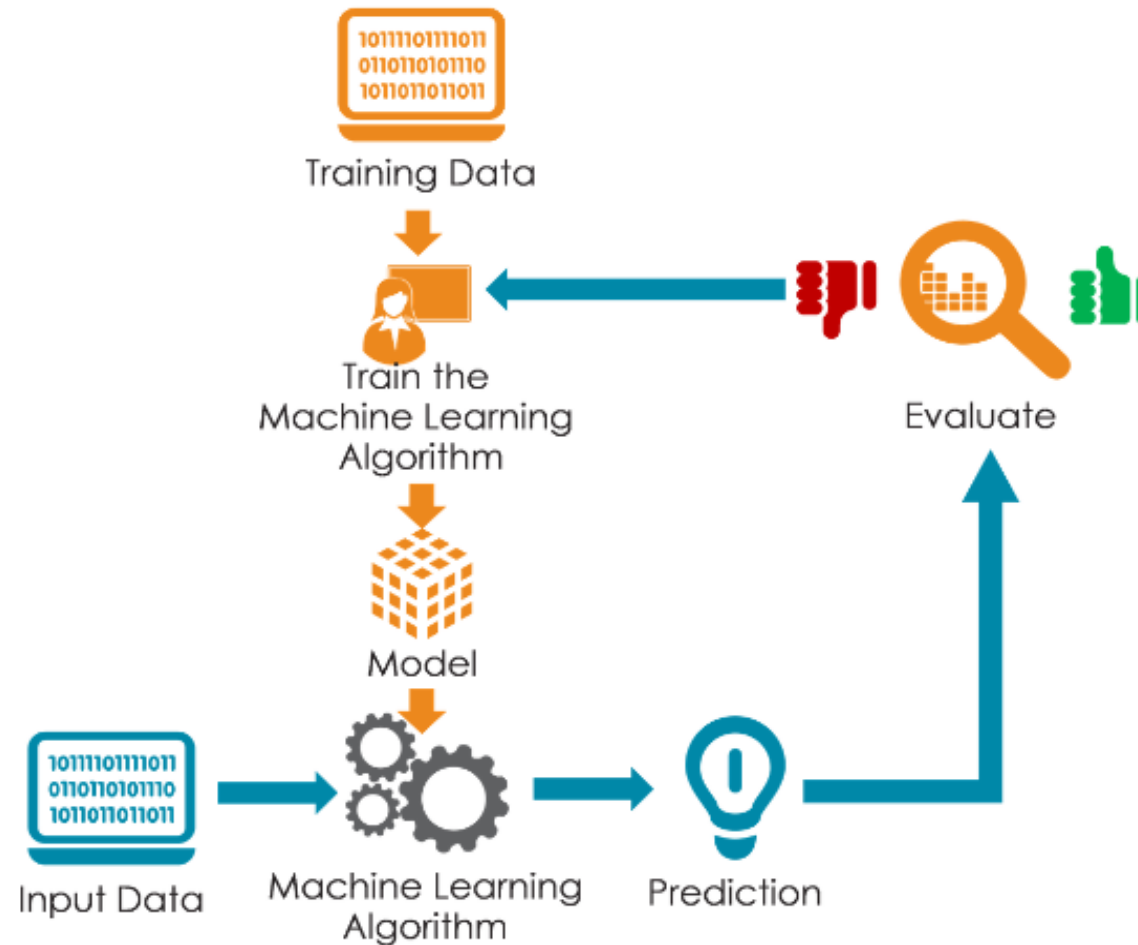


General Pipeline of Machine Learning

- Data representation + Modelling + Evaluation + Optimisation (Pedro Domingos, 2012)





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COMP4139

Machine Learning (MLE)

- Supervised Linear Models

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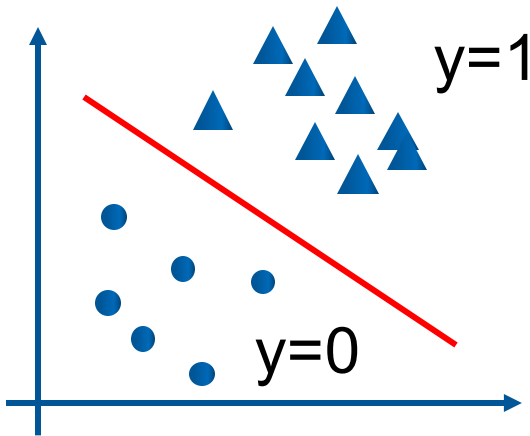


Learning Objective

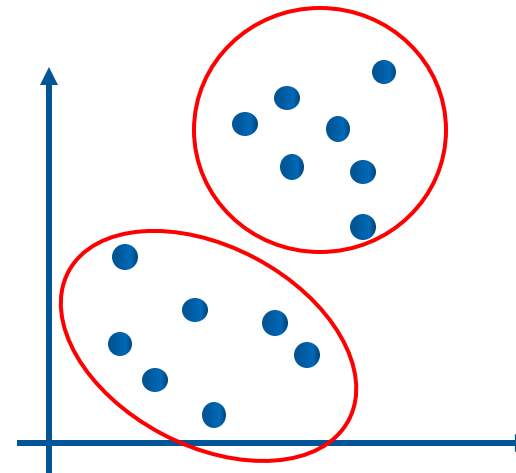
- Concepts about ML
- Univariate/Multivariate Linear Regression
- Logistic Regression
- Naïve Bayes
 - Multinomial Naïve Bayes
 - Gaussian Naïve Bayes

What is supervised learning?

- Supervised vs Unsupervised
 - Both require a training dataset.
 - Supervised learning requires labels y for each instance.
 - Unsupervised learning does not use labels.



Supervised



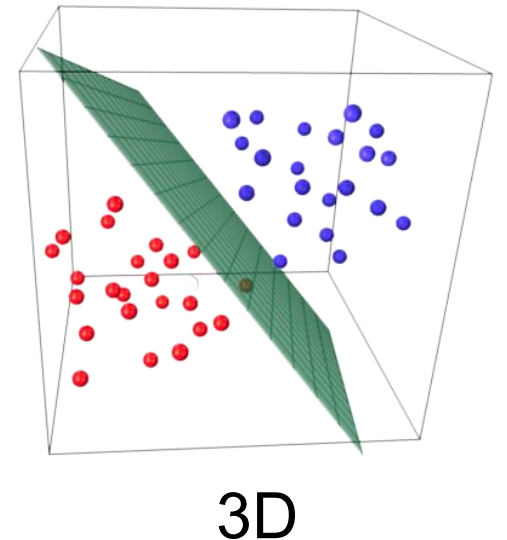
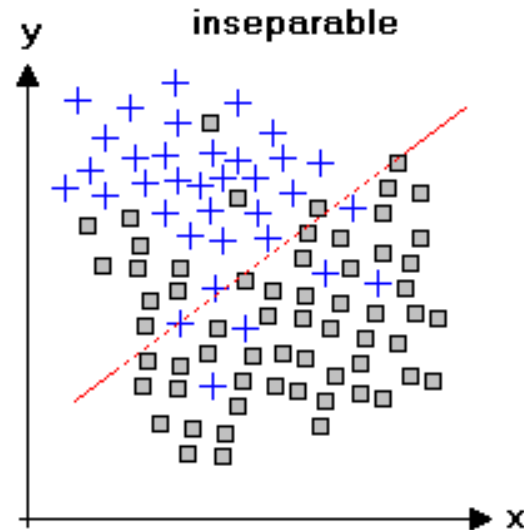
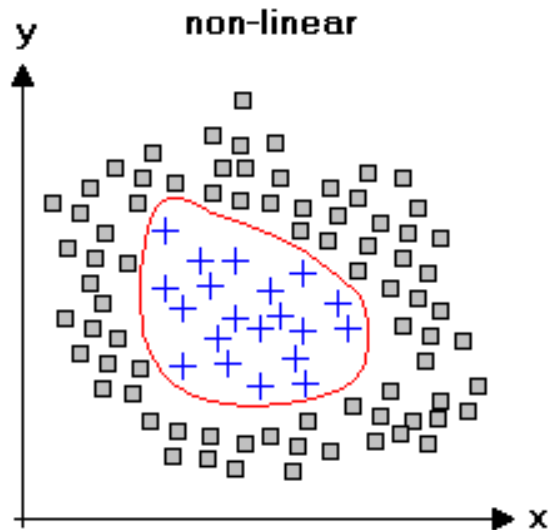
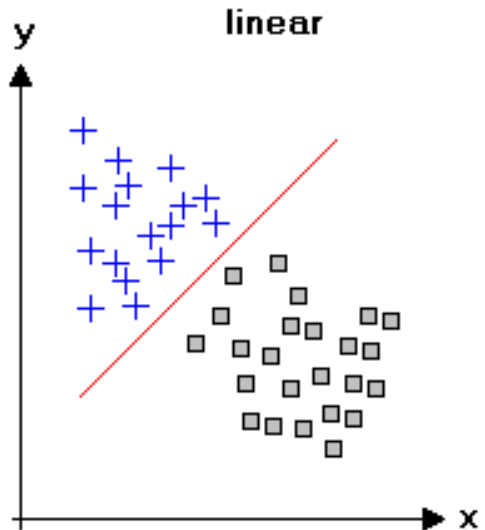
Unsupervised



Linear Model vs Nonlinear Model

- Linear algorithms assume that the sample features x and the label output y are linearly related, described by $f(x)=\langle w,x\rangle+b$. It also often refers to a linear decision boundary in classification models.
 - Linear regression
 - Logistic regression (classification)
 - Naïve Bayes
 - Support Vector Machines (SVM)
- Nonlinear algorithms assume a nonlinear relationship between x and y . Thus, $f(x)$ can be a function of arbitrary complexity.
 - K nearest neighbour
 - Kernel SVM
 - Decision Trees
 - Neural Networks

- Datasets whose classes can be separated by linear decision surfaces (100% classification accuracy in the training set)
- Implies no class-overlap
- Classes can be divided by e.g. lines for 2D data or planes in 3D data





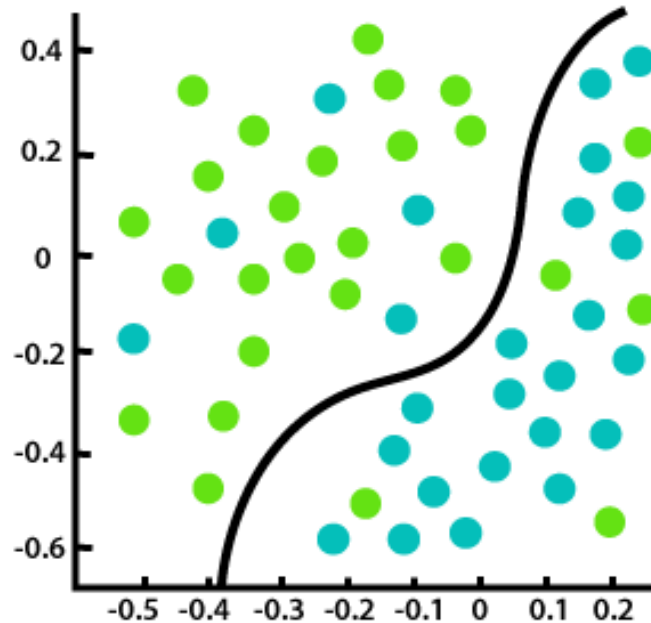
Parametric vs Non-Parametric Algorithms

- Parametric algorithm (**model-driven**): assumes the data follows a specific distribution in its feature space or a pre-defined relationship between feature and outcome.
 - Linear regression
 - Gaussian naïve Bayes
 - Maximum likelihood classifier
- Non-parametric algorithm (**data-driven**): approaches are not constrained to prior assumptions on the data distribution.
 - Decision trees
 - Neural networks

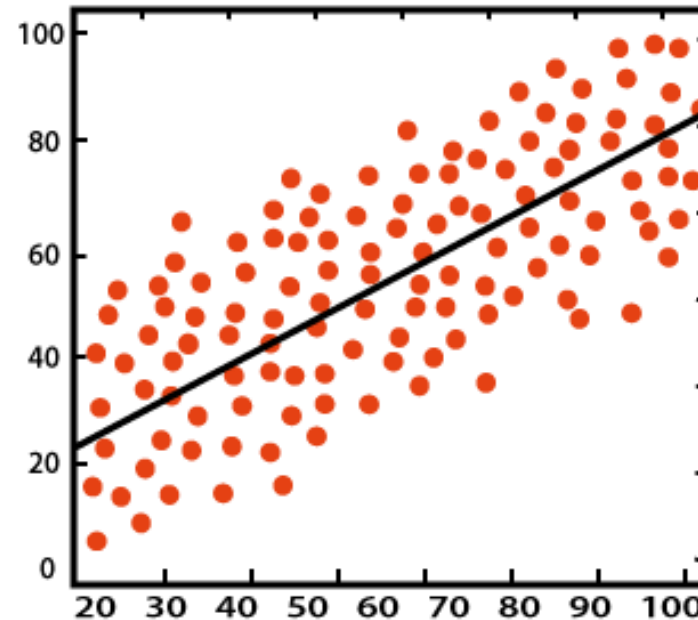


Classification vs Regression

- Regression (predicting numerical values): estimate **values** for a given data.
- Classification (predicting categorical values): assign **class labels** to data.



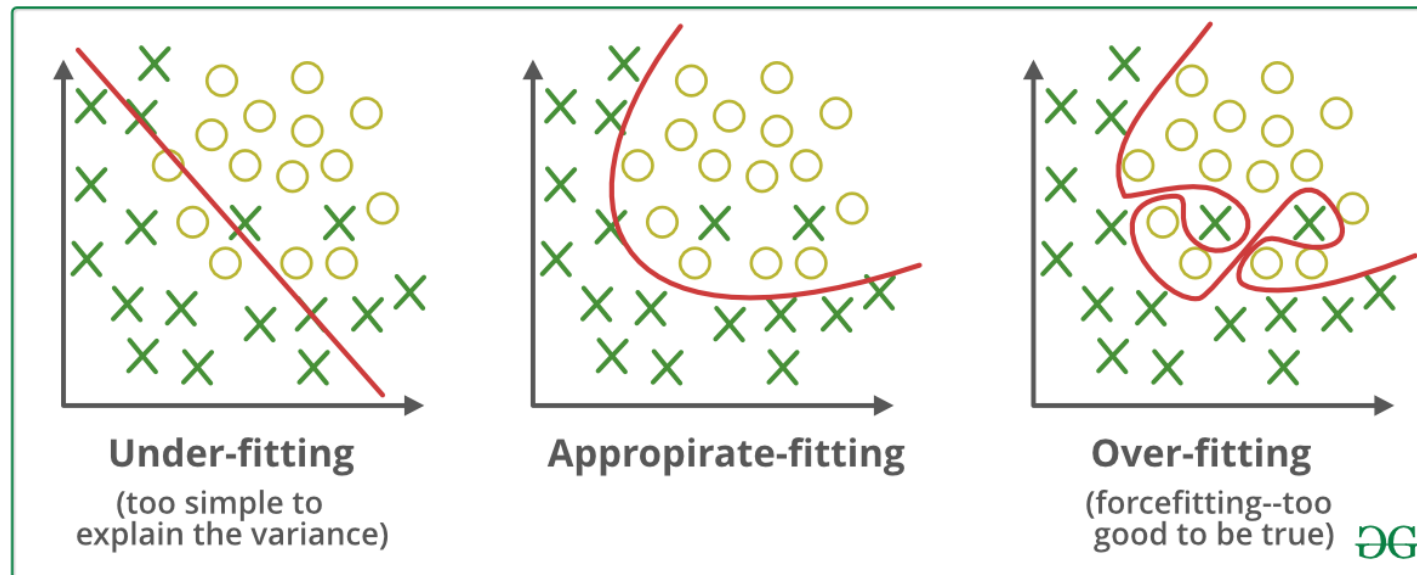
Classification



Regression

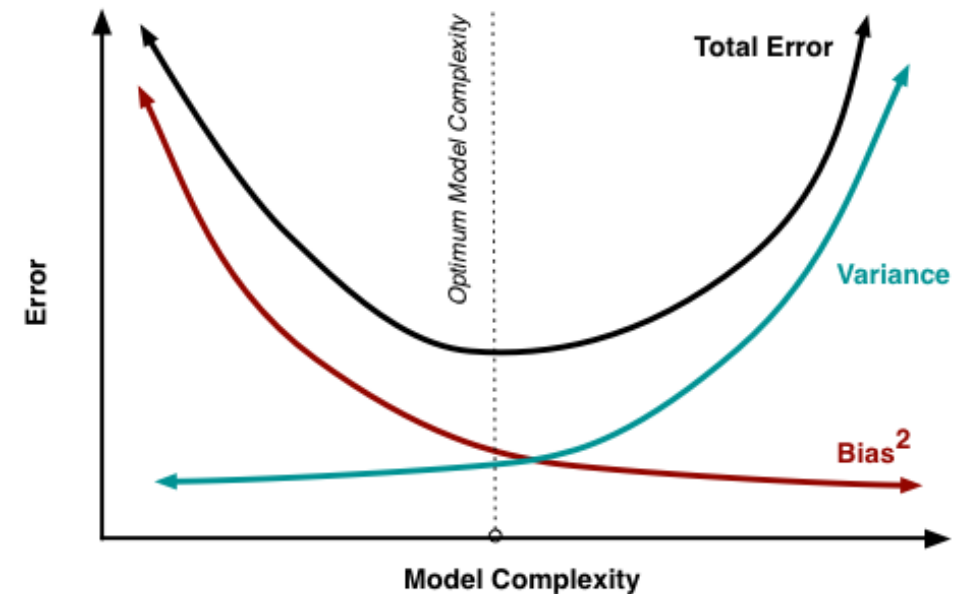
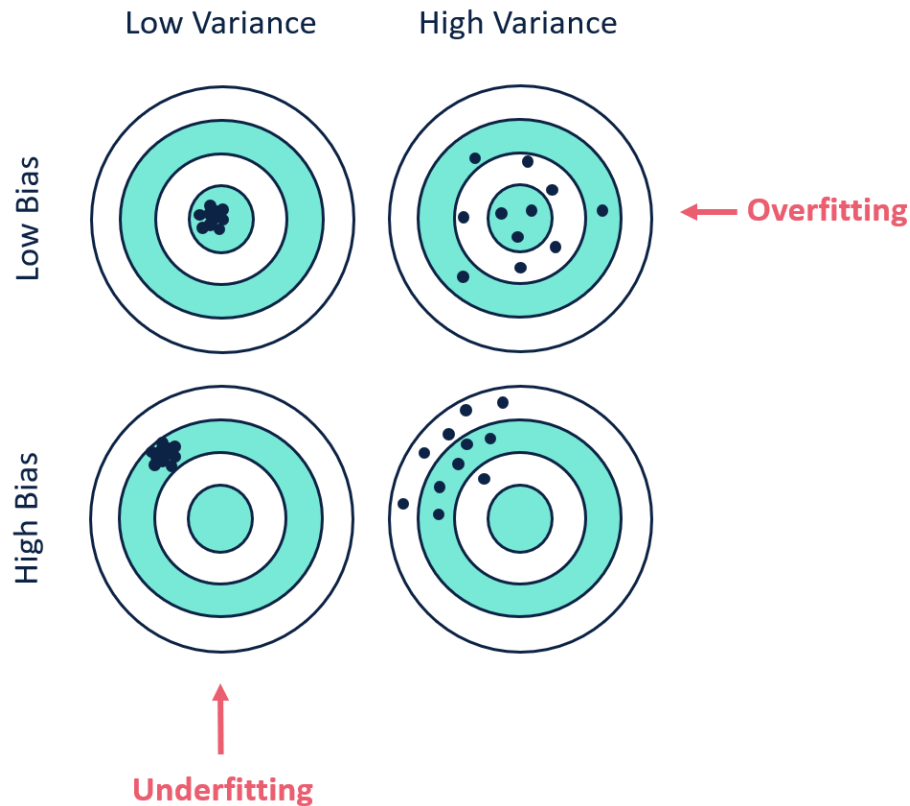
Overfitting vs Underfitting

- Overfitting: a model learns to map the training data too well, which negatively impacts the performance of the model on new, unknown, data. **Poor generalisability.**
- Underfitting: if it neither can model the training nor the test data correctly. Underfitting is easy to detect in the training phase using evaluation metrics. **Poor performance.**



Variance and Bias

- **Bias** indicates the error between the approximated model to the ideal model.
- **Variance** is the amount that the estimate of the target function will change given different training data.





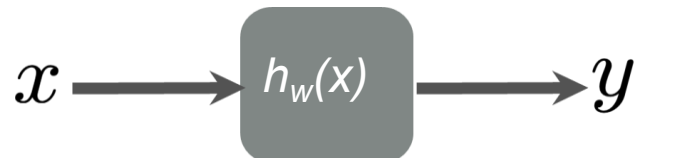
Intrinsic Parameters vs Hyperparameters

- Intrinsic parameters
 - Can be efficiently learned on the training set
 - Large in number
 - e.g. weights in linear regression or Artificial Neural Network (ANN)
- Hyper-parameters
 - Must be learned by establishing generalisation error
 - No efficient search possible
 - Smaller in number
 - e.g. the number of nodes in an ANN or the weights of two terms in a loss function



Univariate Linear Regression

- Univariate Linear Regression determines relationship between **one** independent (feature/predictor: x) variable and one dependent variable (outcome: y).
- w_0 and w_1 are the model parameters to be estimated.
- Train process: given a model h with solution space S and a training set $\{X, Y\}$ that contains n samples, a learning algorithm finds the solution $\hat{S}$ that minimises the cost function $J(S)$


$$h_w(x) = w_0 + w_1 x$$

$$J = \min_{w_0, w_1} \sum_{i=1}^n (h_w(x_i) - y_i)^2$$



Multivariate Linear Regression

- Multivariate linear regression is the linear sum of the multiplications of each of the features with their corresponding weight terms:

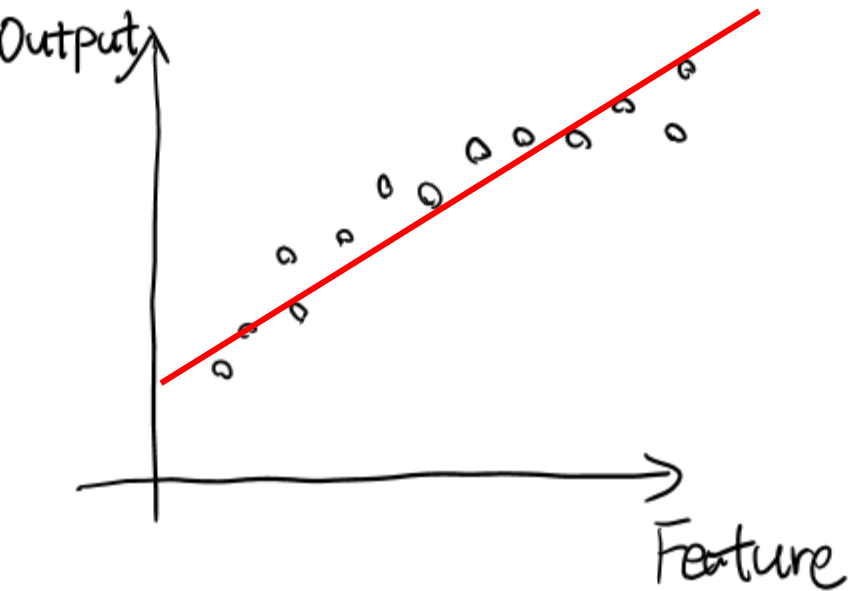
first-order polynomial: $\hat{y} = h(\mathbf{x}, \mathbf{w}) = w_0 + w_1x_1 + \dots + w_jx_j + \dots + w_dx_d$

higher-order polynomial: $h(\mathbf{x}, \mathbf{w}) = w_0 + w_1x_1 + w_2x_1^2 + \dots + w_kx_1^k + \dots + w_{k*d}x_d^k$

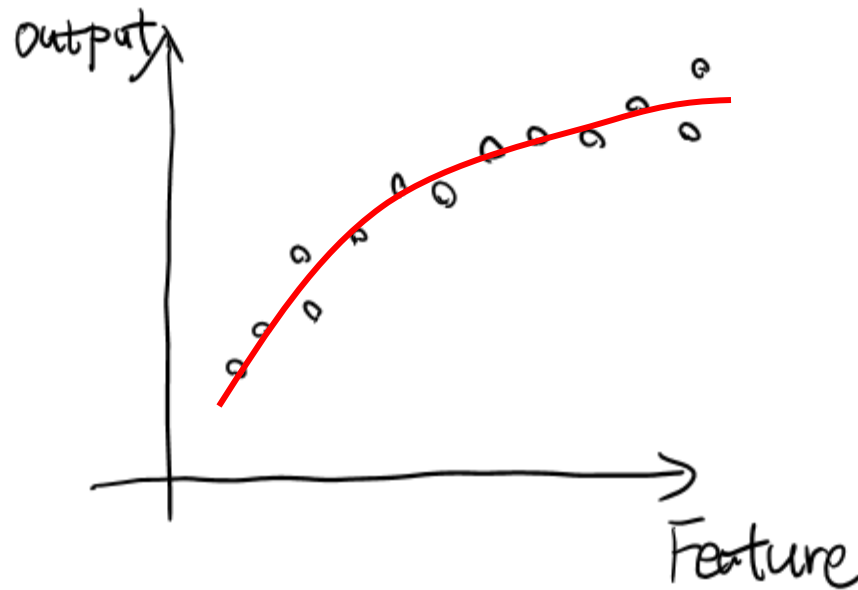


Linear/Polynomial Regression

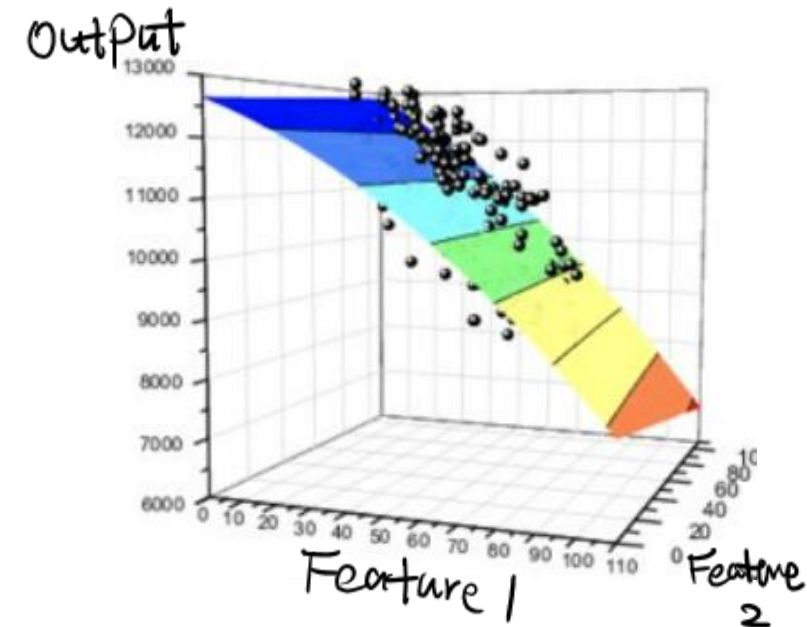
Univariate Linear Regression
(line)



Univariate Polynomial Regression



Multivariate Polynomial Regression





Linear Classification (Logistic Regression)

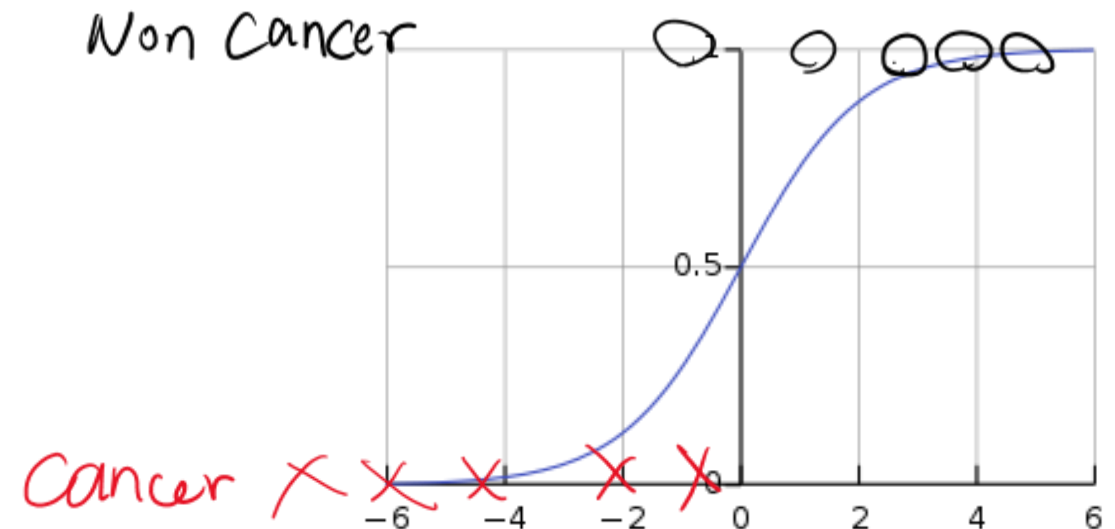
- Logistic regression provides probabilities and classifies new samples using continuous and discrete measurements.
- Output is always between 0 and 1, indicating the likelihood of being one class.
- Use Maximum Likelihood or minimise the Cross Entropy

$$\text{Logistic function } P(x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$$

True label Predicted probability

$$L = \sum_{k=1}^K (y_k \ln(p_k) + (1 - y_k) \ln(1 - p_k))$$

Find the best w_0, w_1 that maximise L or minimise $-L$





Naïve Bayes

- Based on Bayes' theorem $P(A|B) = \frac{P(B|A)P(A)}{P(B)}$
- Find the probability of **A** happening, given that **B** has occurred.
- Assumption is that features are independent with the same importance (Naïve).
- Naïve Bayes leads to a linear decision boundary
- Multinomial Naïve Bayes (features are counts: number of occurrence)
- Gaussian Naïve Bayes (features are continuous values)
- Bernoulli Naïve Bayes (features are only binary values: yes or no)



Multinomial Naïve Bayes

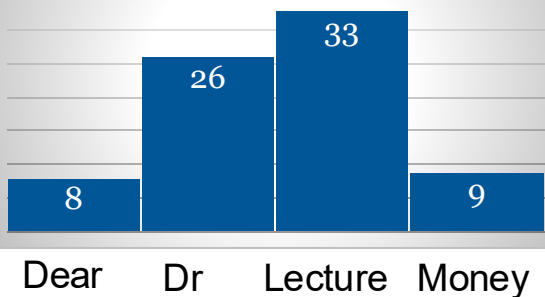
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Application: Spam email detection

Test: **Lecture, Money**

Training Data

Normal



$P(B|A)$

$$\begin{aligned} P(\text{Dear}|N) &= 8/76 = 0.105 \\ P(\text{Dr}|N) &= 26/76 = 0.342 \\ P(\text{Lecture}|N) &= 33/76 = 0.434 \\ P(\text{Money}|N) &= 9/76 = 0.118 \end{aligned}$$

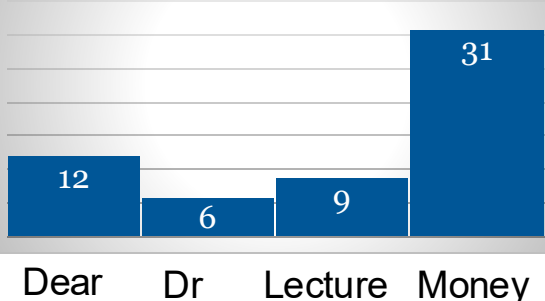
$P(A)$: Prior probability

$$P(N) = 76/(76+58) = 0.57$$

$$P(A|B) \propto P(B|A)P(A)$$

$$\begin{aligned} P(N | \text{Lecture, Money}) \\ &= P(N) * P(\text{Lecture}|N) * P(\text{Money}|N) \\ &= 0.57 * 0.434 * 0.118 \\ &= 0.029 \end{aligned}$$

Spam



$$\begin{aligned} P(\text{Dear}|S) &= 12/58 = 0.207 \\ P(\text{Dr}|S) &= 6/58 = 0.103 \\ P(\text{Lecture}|S) &= 9/58 = 0.155 \\ P(\text{Money}|S) &= 31/58 = 0.534 \end{aligned}$$

$$P(S) = 58/(76+58) = 0.43$$

$$\begin{aligned} P(S | \text{Lecture, Money}) \\ &= P(S) * P(\text{Lecture}|S) * P(\text{Money}|S) \\ &= 0.43 * 0.155 * 0.534 \\ &= 0.036 \end{aligned}$$

0.036 > 0.029, it is a Spam!



Gaussian Naïve Bayes

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

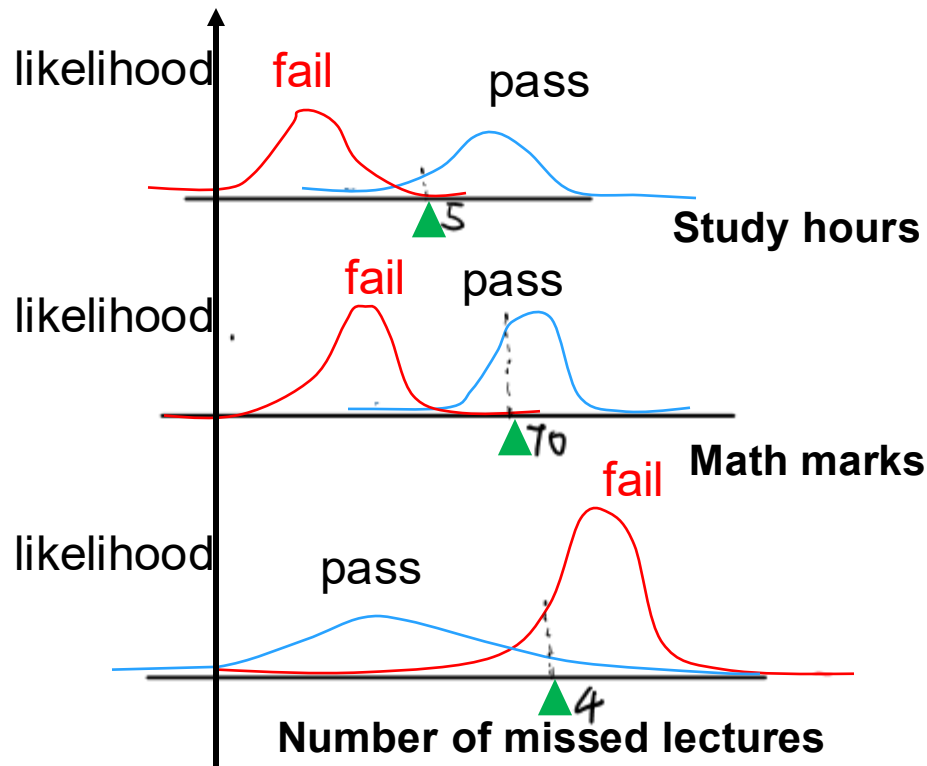
Application: if you'll pass Machine Learning

Training Data: convert data to Gaussian distributions

$P(A)$: Prior probability

Test: study hours (5), math marks (70), missed(4)

$$P(A|B) \propto P(B|A)P(A)$$



$$P(P) = 0.7$$

$$\begin{aligned} P(P|\text{hours}=5, \text{mark}=70, \text{missed}=4) \\ &= P(P) * L(\text{hours}=5|P) * L(\text{mark}=70|P) * L(\text{missed}=4|P) \\ &= 0.7 * 0.03 * 0.08 * 0.01 \end{aligned}$$

We use **log** to avoid underflow:

$$\begin{aligned} &\log(0.7 * 0.03 * 0.08 * 0.01) \\ &= \log(0.7) + \log(0.03) + \log(0.08) + \log(0.01) \\ &= -10.99 \end{aligned}$$

$$P(F) = 0.3$$

$$\begin{aligned} P(F|\text{hours}=5, \text{mark}=70, \text{missed}=4) \\ &= P(F) * L(\text{hours}=5|F) * L(\text{mark}=70|F) * L(\text{missed}=4|F) \\ &= 0.3 * 0.00001 * 0.000001 * 0.04 \end{aligned}$$

$$\begin{aligned} &\log(0.3) + \log(0.00001) + \log(0.000001) + \log(0.04) \\ &= -29.75 \end{aligned}$$

-10.99 > -29.75, it is a Pass!



Learning Objective

- Concepts about ML
- Univariate/Multivariate Linear Regression
- Logistic Regression
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